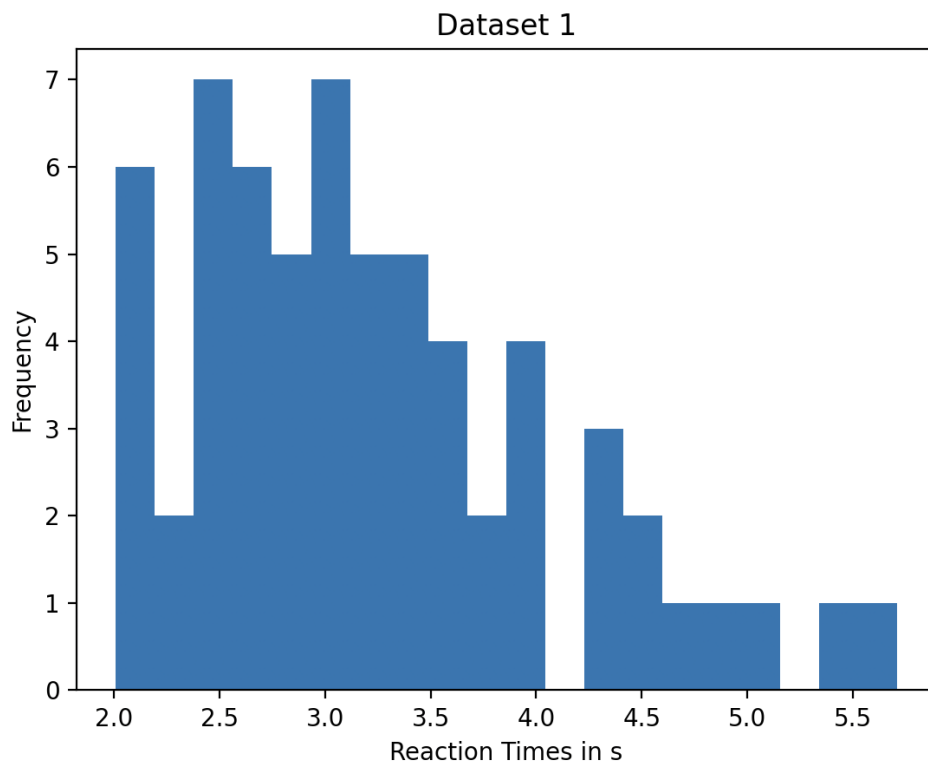


# RMSSD

## Assignment 1

### Dataset 1



### Insights

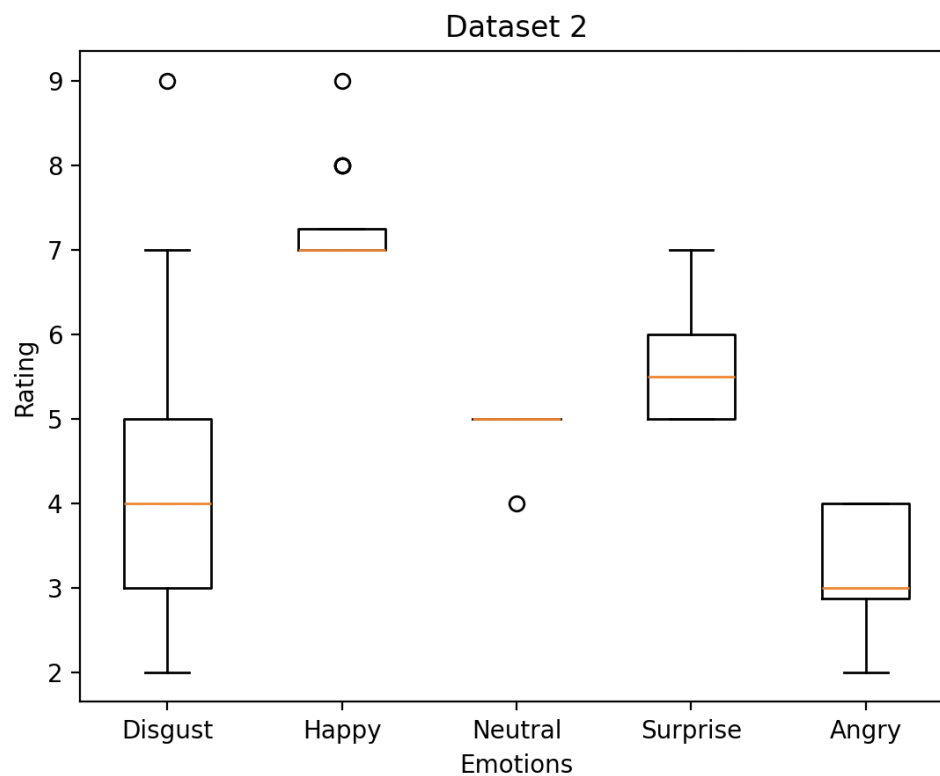
- 1) The graph is positively skewed which means that majority of people in this dataset have reaction times on the lower side (around 2.5 - 3.5 sec).
- 2) There is a drop after the **lowest reaction time** which means that there are only some people with fast reaction times.
- 3) The min and max reaction times are in classes with lower limits approximately 2 and 5.5 seconds respectively.

### Choice of graph

I chose a histogram because we had reaction times with a lot of decimal precision and hence clubbing it in continuous data to study frequency makes sense. Since a histogram shows frequency distribution of continuous data, we can gather a lot of insights into the dataset using this. Also, other types like pie charts, line charts don't make sense here. A

histogram is preferred than box plot because a histogram also emphasizes on the frequency and a box plot will emphasize more on the range, distribution of reaction times.

## Dataset 2



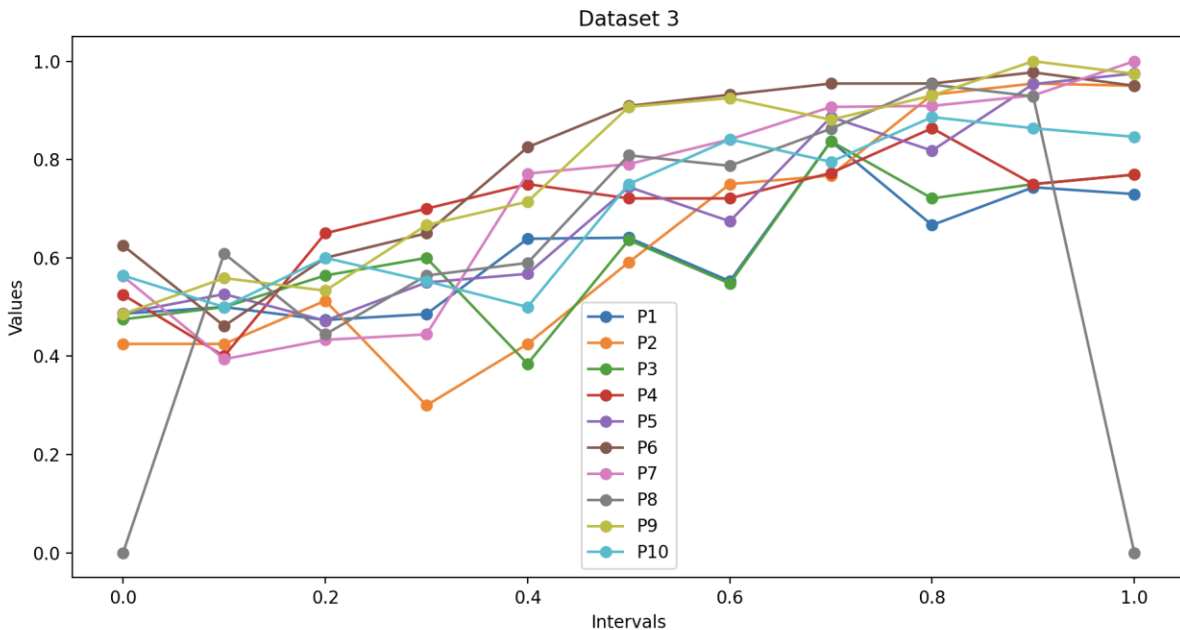
### Insights

- 1) The happy and neutral are mostly same i.e at 7-8 and 5 respectively with a few outliers.
- 2) Disgust, surprise and angry are fairly varying. Disgust is the only one with the most stretched out values. Surprise is medium between 5 and 7. Angry is low between 4 and 2.
- 3) So we can get a kind of average that most models give 5 to neutral, 7-8 for happy, medium values like 6 for surprise and low to angry at around 3. Disgust is varying for all.

### Choice of Graph

I chose box plots each for different emotions since it best depicts the variation of each emotion across different models. Since box plot includes median, quartiles etc it will successfully show the range and variation of each emotion. It is the best to depict the data completely.

## Dataset 3



I have named the column header (0, 0.1, 0.2 ... 1) intervals and the rating of each participant as values since it was not clear in the dataset.

### Insights

1) We can see a basic trend which is roughly followed in all the participants. The line starts from around 0.5 (for interval 0) and rises slightly till around 0.8 (for interval 1). For green (P3) and orange (P2) we can see variation in the middle intervals.

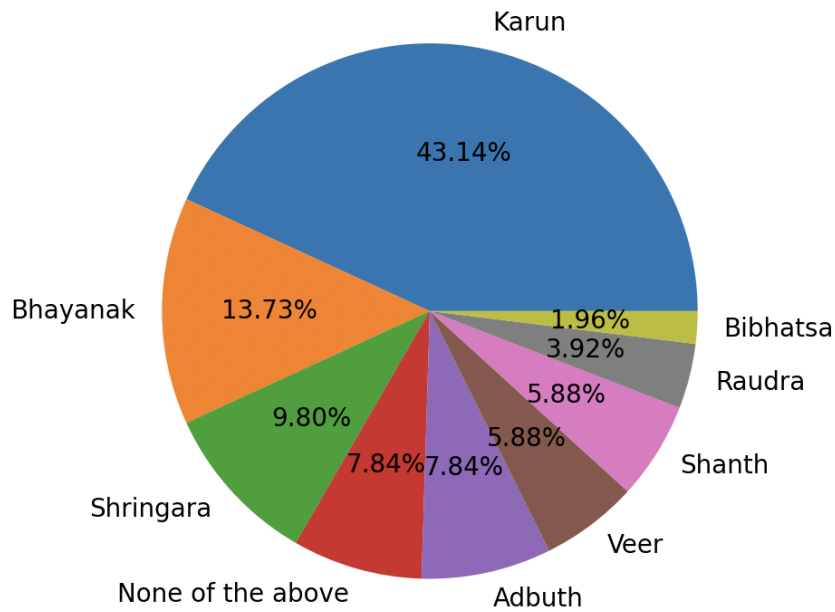
2) Most different values are for P8 for the first and last interval. For all the other intervals the value lies in the colorful band.

### Choice of Graph

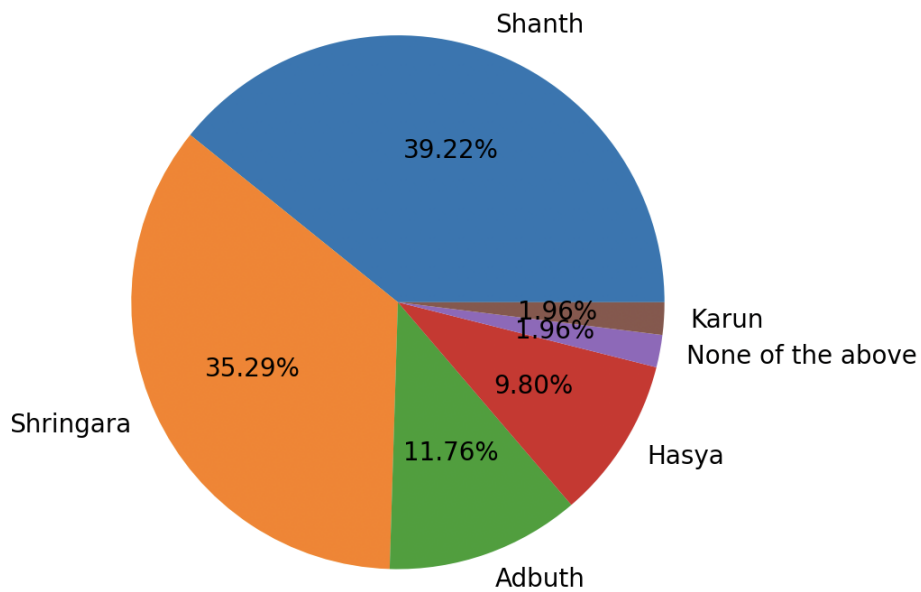
I chose a line chart since it shows how the value varies for each person. It shows the trend in the values with respect to the intervals. Using elimination also any other plot like box, histogram, pie plots would not be helpful to depict the 10 participant's data.

### Dataset 4

Music Piece 1



Music Piece 2



### Insights

- 1) We can clearly see from the piechart that most participants rated music piece 1 as Karun: Sorrow (43%). Some other less significant are Bhayanak: Fear (13%) and

Shringara: Love (9%). So we can say that it is a music of sorrow with maybe components of love and fear.

2) Again from the piechart that most participants rated music piece 2 as Shanth: Peace (39%) and Shringara: Love (35%). So we can say that it is a music of calmness and love.

3) There are no categories like Bhayanak: Fear, Veer: Heroism etc in music piece 2.

### Choice of Graph

Piechart is the choice in the dataset because it clearly shows the percentages (using autopct in matplotlib) of various categories of emotions which help to depict the whole data. Also, other types like histograms, line charts etc don't make sense here.